

Drain Water Heat Recovery Devices

Drain water heat recovery (DWHR) devices are little-known but effective means to save hot water energy and homeowner dollars.

by Charles Zaloum, Maxime Lafrance, and John Gusdorf

Heat recovery ventilators (HRVs) recover heat energy from air exhausted from homes, saving energy and money for the homeowner. But houses can use up to 20% of their energy heating water, and much of that energy can slip away down clothes washer, sink, and shower drains. Drain water heat recovery (DWHR) devices are little-known but effective means to save hot water energy and homeowner dollars.

At the Canadian Centre for Housing Technology, we have been studying the effectiveness of DWHR devices. In our 2005 study, we found that DWHR devices recover energy almost exclusively during simultaneous water draws (showers), and that for modeling purposes, all other water draws could be ignored. Based on this conclusion, we sought to characterize the DWHR devices in order to develop an energy savings calculator. The purpose of our experiment was twofold. Our first purpose was to assess the performance of DWHR devices. Our second purpose was to develop standard tests and modeling methods that would enable manufacturers, utilities, policy makers, and consumers to estimate energy savings caused by these devices.

The DWHR technologies we tested are fairly simple in design and can be effective in reducing the amount of energy needed to produce hot water (see Table 1). The units used in this study are of various lengths and configurations, but they all have the same basic design. This design

consists of 3-inch nominal (76.2-mm) copper drainpipe wrapped with either 1/2-inch nominal (12.7-mm) or 3/8-inch nominal (9.5-mm) soft copper tubing. Cold water is circulated through this copper tubing, recovering heat from the drain.

We performed the experiments using two different flow configurations, three different flow rates, and three different shower temperatures in an effort to assess the performance and heat transfer rate of each unit in comparison with the others. The two different flow configurations change the volumetric flow rate: this in turn affects the heat transfer and performance of the DWHR units.

The performance of each unit was measured in terms of number of heat transfer units (NTU) and unit effectiveness. The NTU is a measure of the heat transfer size of the heat exchanger. The larger the NTU, the more the heat exchanger approaches its thermodynamic limit. The real effectiveness of the DWHR unit, however, can be defined by the ratio of the actual rate of heat transfer to the maximum possible rate of heat transfer.

Excessive pressure loss in the pipes can affect water flow rate and lead to consumer complaints. We measured the static pressure loss across the DWHR units using pressure transducers that were connected to the data logger.



Left to right, the Retherm S3-60, GFX G3-60, and the Power Pipe R3-60 are some of the tested devices.

Test Setup and Procedures

To properly evaluate the various DWHR units, we needed a controlled environment and standardized tests. All tests were performed in identical conditions, at the Canadian Centre for Housing Technology (CCHT). This allowed us to compare performance between units and model performance for a given unit, and to assess experimental repeatability.

We installed the DWHR units vertically on the main drainpipe, which is a 10-foot straight run to the shower drain (see Figure 1). The units were wrapped with 1/2-inch closed cell foam insulation to ensure minimal heat loss or gain to the surroundings and to reduce the risk of surface condensation. Two pressure gauges were mounted on the coldwater piping, one at the inlet and the other at the outlet, of the outer coil. Pressure transducers were later installed to replace the dial gauges, in order to measure the pressure loss in the outer coil more accurately. Thermocouples were installed on the top and bottom of the plastic drainpipes leading to and from the DWHR units.

The water flow configurations were an important aspect of the experiment, as they affect the flow rate going through the heat exchanger. We used two different water flow configurations:

Configuration A: Route all cold water flow going to the hot water tank (HWT) through the DWHR unit. In other words, the unit preheats the water going only to the HWT.

Table 1. Models Tested

Manufacturer	Model	Actual Length (in)	Coiling Length (in)	Drain Pipe Dia. (in)	Tubing Dia(in)
Power Pipe	R3-60	64	55.5	3	0.375
Power Pipe	R3-36	36	31	3	0.375
GFX*	G3-60	65	60.25	3	0.5
GFX	G3-40	40	36	3	0.5
Retherm	C3-40	40	36	3	0.5
Retherm	S3-60	60	28 x 2	3	0.5

*currently distributed under the name "ECO-GFX"

Configuration B: Route all cold water flow going to the HWT and the shower through the DWHR unit. In other words, the unit preheats all the water going to the HWT and shower.

We did not assess a third possible configuration, in which cold water warmed in the DWHR is routed to the cold water shower tap, because resource and time constraints prevented us from doing so.

We tested three different shower flow rates: 6.5 liters (1.7 gallons) per minute, to simulate an ultra low-flow showerhead; 8.5 liters (2.3 gallons) per minute, to simulate a low-flow showerhead; and 10.5 liters (2.6 gallons) per minute, to simulate an older-style high-flow showerhead.

The tests were also performed at three different water temperatures: 37°C (99°F), 41°C (106°F), and 45°C (113°F), to simulate cool, warm, and hot shower temperatures respectively.

The static pressure loss across the DWHR units was measured using pressure transducers that were connected to the data logger. The pressure losses were measured at 6 liters per minute, 8 liters per minute, 10 liters per minute, and 12 liters per minute, and at maximum flow.

Since the city water temperature where the tests took place fluctuates by approximately 13°C over the course of the year, we had to devise a method of controlling cold water temperature. The national average city water temperature in Canada is 8°C (46°F), so the tests needed to be conducted at a constant cold water supply temperature of 8°C. Given the time of year and the length of time needed to perform the testing, the water was warmer than the national average. The solution to the problem involved using a 2kW

chiller. Although this chiller was not capable of handling the direct required cooling capacity at the required flow rate, it could cool a large volume of water over a longer period of time. The chiller was used to chill two reservoirs of 150 liters and 151 liters respectively (about 40 gallons each).

Once the two reservoirs were cooled, in a closed-loop arrangement powered by a circulating pump, they were connected to the house's water supply.

At first, we assumed that we needed to evaluate the effect of all the parameters (three flow rates, three temperatures, and configurations A and B) on performance, with a total of 18 tests per DWHR unit. But after we performed the tests on two of the DWHR devices, we found that shower temperature had no observable effect on developing NTU curves for modeling. As a result, we eliminated the temperature parameter and set the temperature to 41°C, which reduced the number of tests by half (eight to ten tests). We also found that for a given DWHR unit, the only parameter relevant to calculating the NTU for the heat exchanger was the flow rate. This calculation could be done as long as a range of flow rates from roughly 4 to 10.5 liters per minute could be obtained.

Configurations A and B were used to maintain consistency with the experimental setup once the NTU correlation was determined.

A simplified test procedure can be developed in the future by keeping the configuration and shower temperature constant while changing flow rates. At first, we ran the test showers for 30 minutes, but we found that this could be reduced to 15 minutes, because

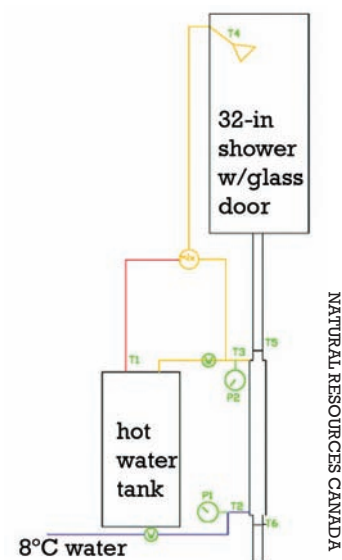


Figure 1. The test setup included pressure transducers (PI, P2) and thermocouples (T1–T6) to measure pressures and temperatures.

steady state was always reached in about 1 minute.

Calculations

Two methods can be used to calculate the performance and the amount of heat transfer for a particular heat exchanger. The first method calculates the logarithmic mean temperature difference (LMTD), and the second calculates NTU effectiveness. We used both methods and compared the results. (For more details refer to the 2005 *ASHRAE Handbook of Fundamentals*, pp. 3-28.)

We concluded that the NTU method was more appropriate, since it does not require exit temperatures. The empirical heat transfer rate coefficient was solved using the actual effectiveness derived from the NTU method.

Using spreadsheet software that permits curve fitting or a numerical method, a standard exponential NTU-versus-flow rate curve can be correlated for each unit, due to the consistent relationship between the NTU and the flow rate. These equations can be used to predict the amount of heat transfer between the fluids, without having to test the unit. All that is required to

work the problem backward to solve the heat transfer rate are the inlet temperatures, the NTU-curves, the flow rate, and the type of unit.

The Bernoulli equation is used to analyze fluid flow from one location to another. In order to use this equation to determine the pressure drop across the DWHR units, four criteria must be met. The fluid must be incompressible, the fluid must maintain a consistent density, the flow must be steady, and the flow must be along a streamline. These conditions are met in this case; therefore, the following equation will be used to determine the flow coefficient to be used in future analysis and/or modeling:

$$\text{Pressure drop (psi)} \Delta P = A \times Q^2$$

Where A is the flow coefficient

Q is the water flow (liters per minute)

Test Results and Analysis

Although the tests were structured in such a way as to ensure repeatability and standardization, some uncontrollable variables—such as ambient temperature, shower water flow rate, and shower temperature—could have affected the results. Since, however, we had determined that each unit's heat transfer performance could be characterized by an NTU-versus-flow rate curve, we knew that the uncontrolled variables had no significant effect. The key variables—flow rate through the DWHR devices, inlet temperature, and outlet temperature—were the important ones. These variables were easily measured and were not affected by external influences.

The performance of each unit was measured in terms of NTU and of the effectiveness, which is the ratio of the actual rate of heat transfer to the maximum possible rate of heat transfer. NTU seemed to show a better correlation factor than did effectiveness.

Analysis was done for both minute 1 through minute 11 and minute 5 through minute 11. The idea was that minutes 1 through 11 would be a more realistic simulation of the entire performance cycle as the heat

exchanger goes through the transient (warm-up) stage, whereas minutes 5 through 11 would simulate a steady-state operation.

NTU

From minute 1 to minute 11, the best-performing unit was the Power Pipe R3-60, followed by the GFX G3-60. The NTU can also be observed on a per foot basis. The GFX G3-40 unit performed best on a per foot basis, followed by the Power Pipe R3-60. The per foot basis may be important from a purely economic standpoint, as copper prices have risen significantly in the last decade (see Figure 2). We observed a tendency for the longer pipes to have a lower NTU per foot. This is because as the two fluid temperatures approach each other, the amount of heat transfer diminishes, which also affects the NTU on a per foot basis.

Effectiveness

The best-performing unit was, again, the Power Pipe R3-60, followed by the GFX G3-60. Effectiveness can also be looked at on a per foot basis. Doing so gives a better understanding of the marginal gain a large pipe offers in comparison to a small pipe (see Figure 3). In this case the GFX G3-40 performed best. The two calculations follow a similar pattern, where the 60-inch pipes have lower values of effectiveness per foot. The reason is that, as the two fluid temperatures approach each other, there is less heat transfer, so a longer pipe becomes only marginally more effective, thus reducing effectiveness on a per foot basis.

Pressure Drop

Static pressure loss across each DWHR unit was measured for each pipe at different flow rates. The GFX G3-60 unit has the greatest pressure loss; this is also the longest unit. The Power Pipe units, on the other hand, have a smaller pressure loss; this is due to their design, in which water travels simultaneously through four $\frac{3}{8}$ -inch tubes, instead of one $\frac{1}{2}$ -inch tube. The Retherm S3-60 also has a low pressure

loss due to its design, which splits the flow into two sections. We found that most shorter units have a similar per foot pressure loss, except for the 36-inch Power Pipe, which has low flow resistance.

DWHR Unit Specifications and Differences

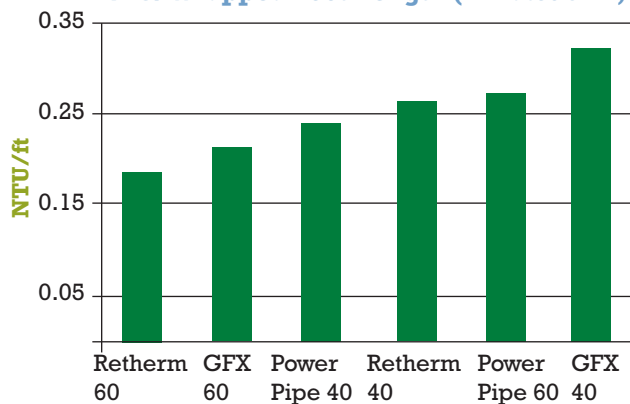
We observed numerous geometrical and design aspects of the eight DWHR units in order to determine if they could be related to performance. We found that the smaller the gap between the tube coiling, the higher the performance; and the lower the vertical flow rate, the higher the performance. (For every rotation around the pipe, the fluid moves vertically by the center-to-center distance between the tubing; the rate at which this movement takes place is the vertical flow rate.)

The Power Pipe R3-60 was one of the best-performing units. This was due to its design, which has four $\frac{3}{8}$ -inch tubes, rather than a single $\frac{1}{2}$ -inch tube, wrapped around the 3-inch pipe. On the other hand, the Power Pipe R3-36 was among the worst-performing units; therefore, the multicoil design may be suitable for long drain pipes only.

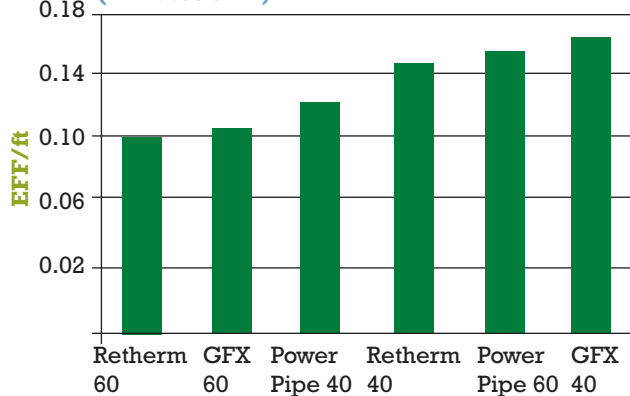
Looking at the GFX G3-40 and the Retherm C3-40, we found that while both designs are virtually identical, the G3-40 outperformed the Retherm C3-40. We believe that the difference may be due to the relative squareness of the tubes on the GFX G3-40. The squarer the tube, the more surface contact area it would have with the pipe, thus increasing the heat transfer area. The GFX G3-40 has a smaller outer circumference, suggesting a possible tighter coil wrapping than that of the Retherm C3-40. If a unit has a larger air gap between the tubing and the pipe, this would affect the heat transfer rate, due to the fact that air is not a good conductor. Ensuring a tighter wrap and a squarer tubing can reduce this effect.

The Retherm S3-60 differed in its design; the cold flow was split into two sections, like two separate units connected in parallel on the same pipe. This didn't appear to improve its performance over that of

NTU/Wrapped Foot Length (Minutes 5–11)



Effectiveness/Wrapped Foot Length (Minutes 5–11)



Figures 2 and 3. The NTU is a measure of the heat transfer size of the heat exchanger. The effectiveness is the ratio of the actual rate of heat transfer to the maximum possible rate of heat transfer.

its counterpart, the Retherm C3-40, which performed almost as well with a single wrap. Splitting the cold flow into two, one flow going to the upper part of the pipe and one to the lower part, would not provide great benefits, as the hot water would have been cooled down by the first section of the heat exchanger before passing to the second. This would reduce the performance of the lower part in terms both of effectiveness and of NTU. (Note, however, that due to its split-flow design, the NTU calculation method presented in this report may not be suitable for the Retherm S3-60.) This design gained an advantage in the pressure loss test, where there was minimal pressure loss over the length of the pipe. In the GFX pipes, the reduced benefit of having a longer pipe could in fact be observed.

The overall performance difference between the GFX G3-60 and the GFX G3-40 was fairly small; with the former

performing slightly better than the latter. But looking at the two on a per foot basis in terms of NTU and effectiveness, the GFX G3-40 is superior to its counterpart. Now, the GFX G3-60 has 60.25 inches of tube wrapping, while the GFX G3-40 has only 36 inches. This indicates that having more tube wrapping confers a marginal benefit. The GFX G3-60 also has a larger outer diameter, which again suggests that the GFX G3-40 could have a tighter wrap. (Note that the GFX G3-60 and GFX G3-40 do not come from the same manufacturer.)

Pressure Loss

The pressure loss data presented in the previous section show that the different pipe configurations can affect the pressure. Pressure loss in the DWHR can become an important factor for low-pressure systems or

rural installations with well pumps. This is where design considerations for longer pipes such as with the Power Pipe R3-60 and Retherm S3-60, discussed above, become important in reducing pressure loss. Shorter pipes would not be expected to cause adverse effects on water pressure. During this experiment, where city water (60–80 psi) was fed to the system, no significant reduction in water flow was observed at the shower.

Full Cycle and Steady State

We analyzed the data for the entire performance cycle, including the transient stage, (minutes 1 through 11) and for the steady state alone (minutes 5 through 11), as described above. We had theorized that a long pipe would take longer to reach steady state than a short pipe. However, when we compared the two sets of results, we found that there was only a slight

difference between the full-cycle data and the steady-state data.

Standardized Testing?

NTU-curves for each pipe must be developed to do energy-saving calculations. A simple test to measure the input-output temperatures at representative flow rates is all that is needed. We found that configurations A and B did not have any effect on developing NTU-curves, and that only the flow rate going through the heat exchanger did. Therefore a standard test can be developed even if the water temperatures are not well controlled; as long as the inlet and outlet temperatures and flow rates are known, the NTU-versus-flow rate curve can be developed.

In this study, a set of nine tests was used to develop an excellent NTU-curve. With curve fit NTU-equations, predicting the behavior of the DWHR units, including the effectiveness and amount of heat transfer, is quite simple.

Using the simple NTU-curves developed as part of this project, and applying the other theoretical calculations presented above, we developed a simple energy savings calculator so that consumers and energy utilities can evaluate the benefits of this technology. The user has to input, or select from a list, information such as shower water temperature, the length of the shower, type of showerhead, water heater, DWHR unit, and so on.

What Works for You?

The best-performing unit was the Power Pipe R3-60, and the best per foot result was achieved with the GFX G3-40 pipe. The unit with the overall least pressure loss was the Power Pipe R3-36. On a per foot basis, the two Power Pipe units and the Retherm S3-60 performed equivalently well in terms of pressure loss, although shorter single-coil units are not expected to cause water pressure problems. We also found that there was an optimal balance between performance and size. As the pipes get longer, they tend

water

Drain Water Heat Recovery Energy Savings Calculator Logo

English ☒ French ☐

Daily Hot Water Usage

Length of showers in minutes: 12

Number of showers per day: 4

Shower temperature: ☐ Cool (37° C) ☒ Warm (41° C) ☐ Hot (45° C)

Type of shower head: ☐ Low Flow (8.5 L/min.) ☒ Standard (9.5 L/min.) ☐ Older (15 L/min.) ☐ High Flow (18 L/min.)

Type of water tank: ☐ Standard Nat. Gas (78%) ☐ High Efficiency Nat. Gas (90%) ☐ Oil (78%) ☒ Electric (100%)

Type of DWHR unit: ☐ R3-60 ☐ R3-36 ☐ G3-60 ☒ G3-40 ☐ C3-40 ☐ S3-60 Manufacturer: GFX

Configuration: ☐ Preheat cold water to hot water tank only ☒ Preheat cold water to hot water tank and shower

Nearest City: Ottawa, ON Click on a red dot on the map below to select a location.

RESULT: 2706.92445 kWh of electricity Reset to defaults



The Web calculator can help you choose which product is most appropriate, or cost-effective, for your customer's or your family's showering habits.

to increase performance only slightly; shorter pipes perform best on a per foot basis. The Web calculator can help you choose which product is most appropriate, or cost-effective, for your customer's or your family's showering habits.

This project only tested a sample of the units available on the market at the time of testing, and manufacturers have developed new, better performing designs. Natural Resources Canada has since set up an independent test facility at the Saskatchewan Research Council that will enable manufacturers to obtain NTU curves for their whole suite of products. The Web calculator will be updated as new units are tested.

Charles Zaloum, Maxime Lafrance, and John Gusdorf all work for Natural Resources Canada and perform research projects at the Canadian Centre for Housing Technology.

For more information:

The Drain Water Heat Recovery-Energy Savings Calculator is available on the Web at: www.ceatech.ca/calculator.

For information about Natural Resources Canada, Sustainable Buildings and Communities, go to www.sbc.nrcan.gc.ca.

For the latest on the DWHR devices tested, go to www.renewability.com (Power Pipe); www.gfxstar.ca (ECO-GFX); and www.retherm.com (Retherm).

For more on the Canadian Centre for Housing Technology, go to www.ccht-cctr.gc.ca.

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